ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C1 · STATES OF MATTER

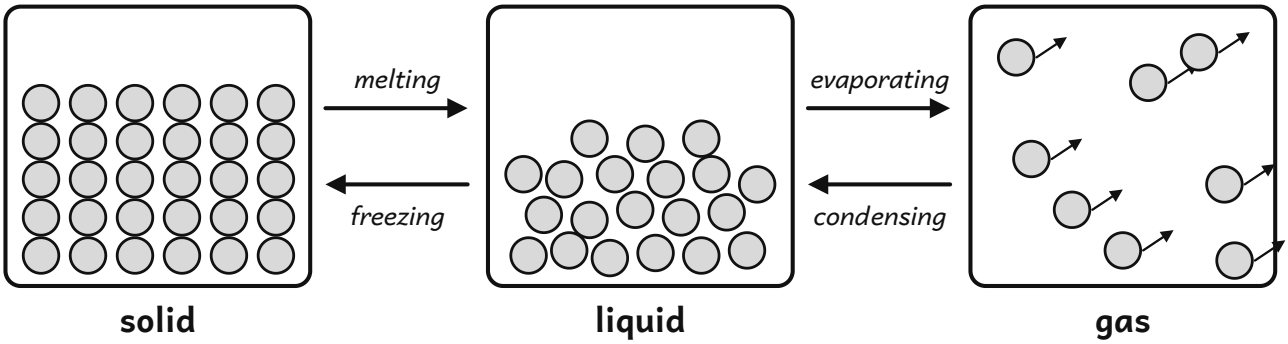
Changes of state

★☆☆
Grade 5

Learning goals

- Name the changes of state.
- Recall key temperatures.

1 **Label the changes of state on the arrows.** [4]



Change	Name
solid → liquid	melting
liquid → solid	freezing
liquid → gas	evaporating / boiling
gas → liquid	condensing

2 **Fill in the gaps.** [5]

WORD BANK

100 **50** **evaporates** **condenses** **salt** **melts** **0**

Water boils at 100 °C. Ice melts at 0 °C. When water vapour touches a cold window it condenses. Puddles dry up because the water evaporates. When salty water evaporates, the salt is left behind.

3 Match each everyday event to the change of state. [5]

Draw a line to match each pair, or write the letter.

wet clothes drying	→ E	**A** melting
dew on the grass in the morning	→ D	**B** boiling
ice cream dripping	→ A	**C** freezing
a kettle making steam	→ B	**D** condensation
water turning into ice cubes in a freezer	→ C	**E** evaporation

4 True or false? [3]

Statement	True	False	If false, correct it
1. Evaporation only happens at 100 °C.		✔	Evaporation happens at any temperature.
2. Condensation is the reverse of evaporation.	✔		
3. The temperature of boiling water keeps rising while it boils.		✔	It stays at 100 °C while it boils.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C1 · STATES OF MATTER

Explaining changes of state

★★☆
Grade 5

Learning goals

- Explain changes using heating and cooling.
- Interpret a heating graph.

1 **Explain each observation.** [6]

a) A cold bottle of water from the fridge becomes wet on the outside.

Water vapour in the air touches the cold bottle, cools and condenses into droplets.

b) Clothes dry faster when they are spread out than when they are folded.

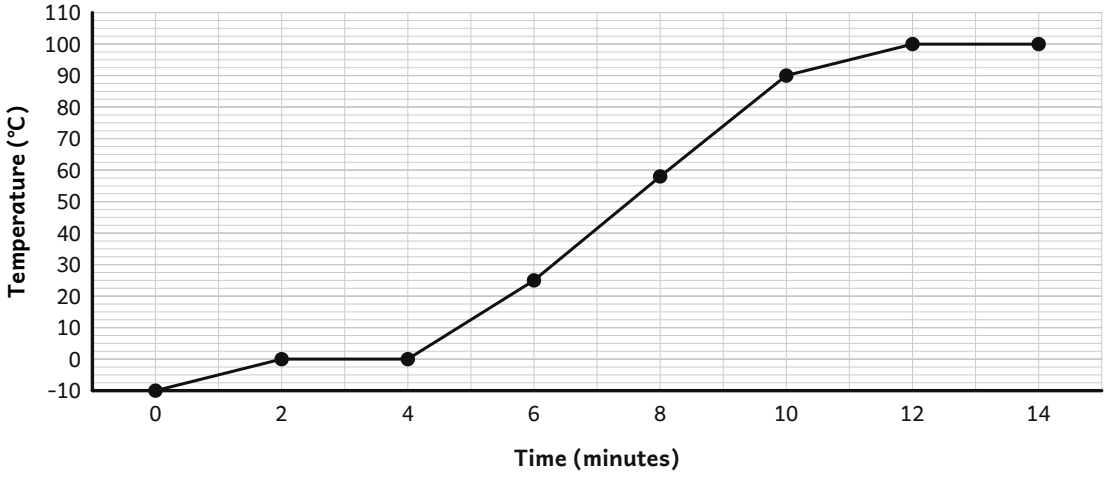
A larger surface area means more water can evaporate at the same time.

c) In winter, people's breath looks like a little cloud.

Warm water vapour in the breath cools quickly in the cold air and condenses into tiny droplets.

2 **Class 5A heated ice and recorded the temperature every minute.** [6]

Time (min)	0	2	4	6	8	10	12	14
Temp (°C)	−10	0	0	25	58	90	100	100



a) Plot the line graph.

b) Why did the temperature stay at 0 °C between 2 and 4 minutes?

The ice was melting: the heat was being used to change solid to liquid.

c) What was happening at 12–14 minutes?

The water was boiling; the temperature stayed at 100 °C.

3 **Why do people in Bangladesh store drinking water in clay pots (*kolshi*) to keep it cool?** [2]

Water seeps through the clay and evaporates from the outside; evaporation takes heat from the pot, cooling the water inside.

ANSWER KEY

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LEARN · PLAY · WONDER

INVESTIGATE · C1 · STATES OF MATTER

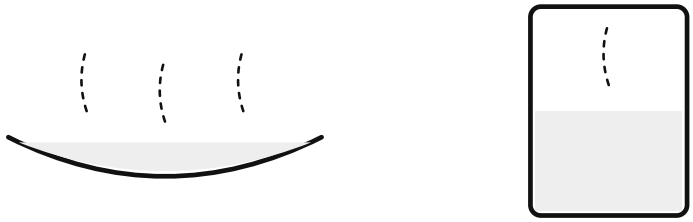
Does surface area affect evaporation?

★★☆
Grade 5

Learning goals

- Plan a fair test.
- Plot two lines on one graph.
- Draw a conclusion.

same volume of water: 100 ml



wide dish **narrow cup**

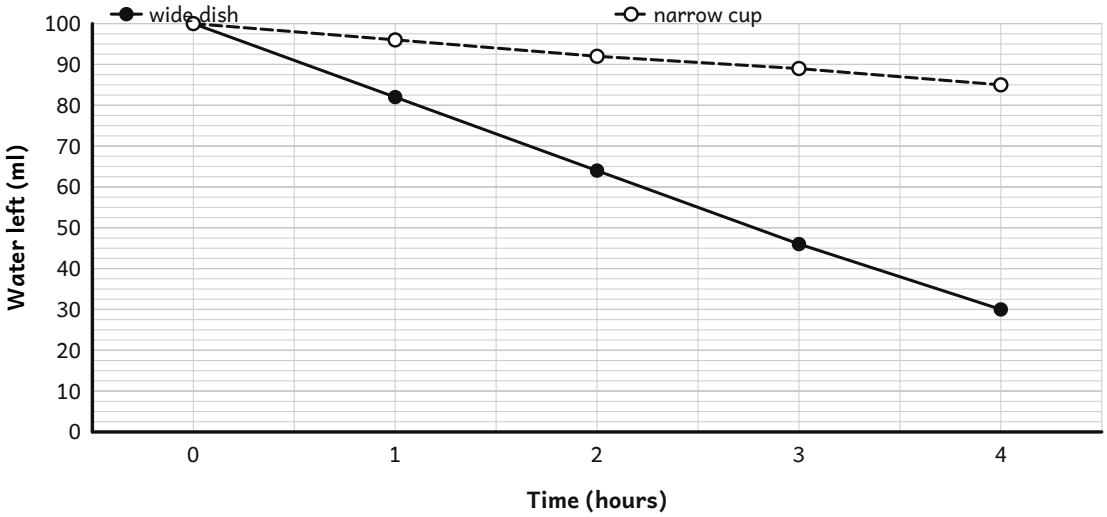
Arif and Mitu put 100 ml of water in a wide dish and 100 ml in a narrow cup. They left both in the same sunny place and measured the water left every hour.

1 **Complete the plan.** [5]

Question	Does surface area affect how fast water evaporates?
Change	the container (wide dish or narrow cup) = surface area
Measure	volume of water left every hour (ml)
Keep the same	starting volume, place, temperature, time, type of water
Prediction	The wide dish will lose water faster because more surface is open to the air.

2 **Plot both sets of results on one line graph.** [5]

Time (hours)	0	1	2	3	4
Wide dish (ml)	100	82	64	46	30
Narrow cup (ml)	100	96	92	89	85



3 **How much water evaporated from each container in 4 hours?** [2]

Dish: 100 – 30 = 70 ml. Cup: 100 – 85 = 15 ml.

4 **Write a conclusion.** [2]

The larger the surface area, the faster water evaporates.

ANSWER KEY

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REAL-WORLD CASE STUDY · C1 · STATES OF MATTER

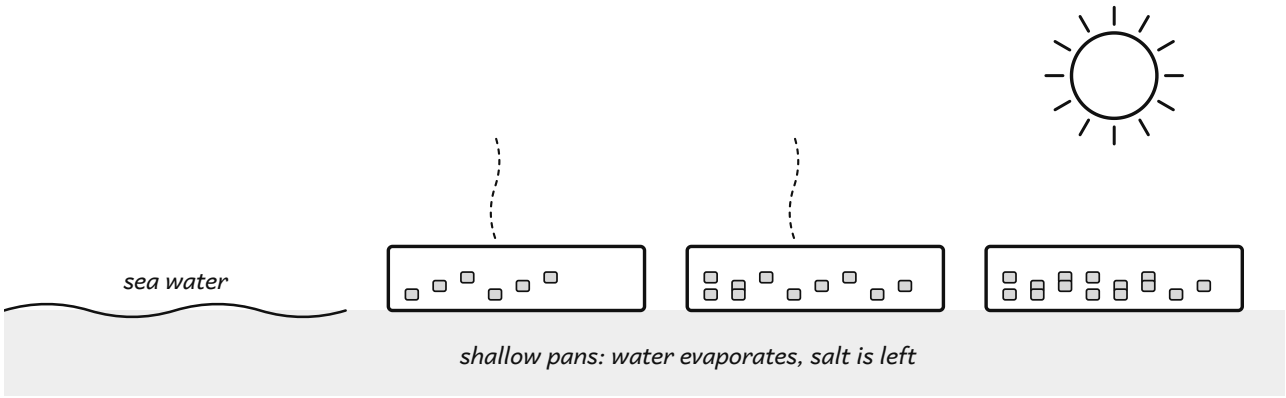
Salt from the sun: Cox's Bazar salt farmers

★★★ Grade 5

Learning goals

- Apply evaporation to a real industry.
- Use data and maths.
- Consider how weather and climate affect people.

Science · Geography · Maths · Economics · Health



Salt pans: sea water is let into shallow fields and the Sun evaporates the water.

Almost all the salt eaten in Bangladesh is made on the coast of **Cox's Bazar** and Chattogram. From about November to May, the dry season, farmers let sea water into shallow, flat fields covered with plastic sheets. The Sun and the wind **evaporate** the water, and white **salt crystals** are left behind. Farmers rake up the salt and carry it away in baskets.

In the monsoon, rain stops salt making: rain dissolves the salt and adds fresh water to the fields. Many salt farmers then grow shrimp or rice instead. Before it is sold, salt is cleaned and **iodine** is added, because our bodies need a small amount of iodine for healthy growth and brain development.

Based on descriptions of the Bangladesh salt industry (BSCIC and news reports). Numbers below are examples.

1 **Explain how salt is made using the words *evaporate*, *dissolved* and *crystals*.** [3]

Salt is dissolved in sea water. The Sun and wind make the water evaporate, and the salt is left behind as crystals.

2 **Why can salt not be made during the monsoon?** [2]

Rain adds fresh water and dissolves the salt; there is less sunshine for evaporation.

3 **About 3.5 kg of salt is dissolved in every 100 litres of sea water.** [3]

a) How much salt is in 1000 litres of sea water?

3.5 × 10 = 35 kg

b) A farmer's field holds 20 000 litres. How much salt could it make?

3.5 × 200 = 700 kg

4 **Why is iodine added to salt?** [1]

The body needs iodine for growth and brain development; everyone eats salt, so it is an easy way to get enough.

5 **Rising sea levels and stronger storms may affect salt farms. Suggest one problem and one solution.** [2]

Problem: storm surges and flooding destroy fields and stop production. Solution: higher earth walls (embankments), early warnings, insurance.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C1 · STATES OF MATTER

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

J	W	S	R	B	D	J	C	J	H
G	A	S	T	V	A	P	O	U	R
M	F	S	M	E	V	S	N	M	E
E	Y	O	Y	V	A	K	D	Y	B
L	F	L	L	H	J	M	E	M	H
T	R	I	P	I	H	G	N	N	E
U	E	D	F	P	Q	O	S	P	G
I	E	A	G	K	B	U	E	H	J
R	Z	I	B	M	E	I	I	T	W
V	E	W	U	B	O	I	L	D	L

boil **condense** **freeze** **gas** **liquid** **melt** **solid** **steam** **vapour**

2 **At what temperature does pure water boil?** [1]

- ○ 0 °C
- ○ 37 °C
- ● **100 °C**
- ○ 212 °C

3 **Which change happens when dew forms?** [1]

- ○ evaporation
- ● **condensation**
- ○ melting
- ○ boiling

4 **Draw the particles in *a* solid, *a* liquid and *a* gas.** [3]

solid
Particles packed tightly in rows.

liquid
Particles close but jumbled, touching.

gas
Particles far apart, moving in all directions.

5 **Challenge. Explain the difference between evaporation and boiling.** [2]

Evaporation happens only at the surface and at any temperature; boiling happens throughout the liquid, with bubbles, at the boiling point (100 °C for water).